

TULSA INTL, OK, USA

WMO: 723560

Lat: 36.199N

Lon: 95.887W

Elev: 198

StdP: 98.97

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13968

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -10.2	(c) -7.6	(d) -16.9	(e) 0.9	(f) -8.3	(g) -14.5	(h) 1.1	(i) -5.5	(j) 11.9	(k) 10.1	(l) 11.1	(m) 8.9	(n) 4.2	(o) 350	(p) 0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	37.8	24.2	36.2	24.5	34.8	24.4	26.5	33.7	25.8	32.9	25.3	32.3	4.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.5	20.0	30.1	23.9	19.2	29.6	23.3	18.5	29.0	83.5	33.7	80.8	33.1	78.4	32.2	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.1	9.8	8.7	-14.5	39.8	3.1	2.1	-16.8	41.3	-18.6	42.5	-20.4	43.7	-22.6	45.3
			-15.2	27.3	3.0	0.8	-17.3	27.9	-19.0	28.4	-20.7	28.8	-22.8	29.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.4	3.7	6.0	11.1	16.1	21.0	25.9	28.5	28.0	23.4	17.0	10.6	5.0
	DBStd	9.95	6.12	6.35	6.04	4.82	4.14	3.10	3.03	3.12	4.18	4.89	5.32	5.58
	HDD10.0	651	210	141	59	8	1	0	0	0	0	6	56	171
	HDD18.3	1895	454	347	234	97	22	1	0	0	8	84	236	413
	CDD10.0	2993	14	28	92	191	342	477	573	558	403	223	74	17
	CDD18.3	1196	0	1	8	30	106	227	314	300	161	43	4	1
	CDH23.3	13673	1	11	73	264	946	2549	4063	3718	1652	364	32	1
CDH26.7	6532	0	2	13	55	297	1159	2198	1987	718	99	3	0	
Wind	WSAvg	4.2	4.2	4.3	4.8	5.0	4.4	4.2	3.7	3.4	3.5	3.9	4.2	4.1
Precipitation	PrecAvg	1028	38	46	81	102	143	117	88	81	107	92	70	60
	PrecMax	1775	95	146	303	265	375	250	289	223	478	237	185	221
	PrecMin	682	0	4	2	9	30	14	2	0	3	5	1	5
	PrecStd	238	27	32	53	54	74	61	72	49	83	61	53	48
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.2	24.9	28.3	30.6	33.2	36.9	39.9	40.6	37.9	32.0	26.5	22.1
		MCWB	13.2	15.7	17.6	19.1	22.7	24.4	23.9	23.7	23.2	22.1	17.8	16.6
	2%	DB	19.0	21.6	25.2	28.2	31.4	35.0	37.8	38.0	34.6	29.4	23.8	19.0
		MCWB	12.5	14.1	16.0	18.8	22.3	24.5	24.4	24.1	23.6	20.7	16.3	13.6
	5%	DB	16.3	19.0	23.0	26.4	29.9	33.6	36.2	36.4	32.8	27.2	21.5	16.4
		MCWB	11.0	12.8	15.4	18.0	21.9	24.1	24.8	24.4	23.3	19.2	15.7	11.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	13.4	16.4	20.7	24.5	28.2	32.2	34.8	34.7	30.9	25.2	19.4	13.8
		MCWB	8.4	11.0	14.5	17.3	21.2	23.8	24.8	24.2	22.4	18.6	14.1	9.6
	0.4%	WB	16.5	18.2	19.5	22.2	24.7	26.9	27.3	27.1	25.7	23.7	20.1	18.0
		MCDB	19.8	22.0	25.5	26.9	30.2	33.4	34.3	34.2	32.5	29.4	23.9	21.1
	2%	WB	14.1	16.0	18.1	20.8	23.8	25.9	26.6	26.2	24.9	22.5	18.1	15.1
		MCDB	16.9	19.5	23.2	25.3	29.1	32.4	34.0	33.4	31.5	27.0	21.6	17.9
Mean Daily Temperature Range	5% DB	WB	11.4	13.9	16.8	19.9	22.9	25.2	26.0	25.5	24.1	21.4	16.7	12.7
		MCDB	15.5	18.1	21.5	24.4	28.3	31.6	33.2	32.9	30.4	25.1	20.2	15.1
		WB	8.6	11.0	15.3	18.6	22.1	24.5	25.5	25.0	23.4	19.9	15.0	10.1
	5% WB	MCDB	12.9	15.7	19.9	23.1	27.1	30.5	32.5	32.1	29.1	23.8	18.7	13.3
Clear-Sky Solar Irradiance	taub	0.302	0.309	0.354	0.387	0.426	0.433	0.443	0.444	0.396	0.351	0.317	0.298	
	taud	2.488	2.484	2.391	2.316	2.268	2.279	2.294	2.255	2.363	2.470	2.510	2.515	
	Ebn at Noon	899	934	913	896	861	852	840	832	858	872	874	881	
	Edh at Noon	82	93	112	127	135	134	131	134	114	93	80	75	
All-Sky Solar Radiation	RadAvg	2.56	3.15	4.14	5.18	5.72	6.55	6.59	5.96	4.98	3.72	2.73	2.17	
	RadStd	0.25	0.47	0.25	0.33	0.54	0.62	0.39	0.40	0.41	0.46	0.33	0.26	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+84
(47) Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page